SHAH MANAV AMIT

SECOND-YEAR B.TECH | ARTIFICIAL INTELLIGENCE & DATA SCIENCE

Mumbai, India | +91 99306 44342 | shahmanavamit@gmail.com

GitHub: shahmanavamit-lang | LinkedIn: manav-shah-b07424380

AI & Data Science student building practical systems in retrieval-augmented generation, algorithmic routing, and data analysis. Strengthening DSA and competitive programming through LeetCode and practice, with hackathon collaboration and independent AI research.

EDUCATION

Shah & Anchor Kutchhi Engineering College, Mumbai | 2025-2029

B.Tech in Artificial Intelligence & Data Science | Second year | **First-year CGPA: 9.08**

TECHNICAL SKILLS

Programming: C++, Python, Java, C, JavaScript, HTML, CSS; FastAPI, Streamlit, Tkinter, Git.

DSA & competitive programming: Data structures, algorithmic problem-solving, graph routing, Dijkstra's algorithm, custom queues and priority queues; ongoing LeetCode / CP practice.

AI / ML & data: RAG, hybrid retrieval, FAISS, BM25, citation verification, Ollama, scikit-learn, feature engineering, NumPy, pandas, model evaluation.

Database foundations: DBMS fundamentals; structured records, data processing, CSV workflows, and analytical reporting.

SELECTED PROJECTS

RAG Citation System | GitHub

Built a PDF Q&A pipeline combining BM25 and FAISS retrieval, grounded generation, and claim-level citation checks in Streamlit. Reported 80% citation faithfulness (8/10 answerable items) on a 14-question evaluation; documented remaining limitations.

S.A.V.E. - Smart Ambulance Vectoring Engine | GitHub

Implemented a three-unit ambulance-routing simulation using Dijkstra's algorithm, road blocking, triage priorities, and critical-call rerouting. Used custom queues and a priority queue without the C++ STL.

Mumbai Pravaas | GitHub

Developed a Mumbai logistics simulation with a 50-vehicle fleet, staged dispatch, priority handling, and traffic-aware routing integrations using OSRM and TomTom.

Additional work: HisaabAI / Smart Expense Analyzer - Python, pandas, Tkinter, and CSV reporting; AI Parking System - synthetic-demand ML prototype with capacity rules.

RESEARCH

RAG-Based Citation-Grounded Document Q&A System - Zenodo preprint, September 2026. DOI: 10.5281/zenodo.22685189

An Ethical Domain-Specific Generative AI Framework for Adolescent Mental Health Triage in India - Conceptual framework, May 2026. DOI: 10.5281/zenodo.20058634

CERTIFICATIONS & PARTICIPATION

Introduction to Model Context Protocol - Anthropic (2026) | MATLAB Onramp - MathWorks (2026) | Yuva AI for All - NASSCOM FutureSkills Prime & INDIAai (2026) | C Programming Training - IIT Bombay Spoken Tutorial (2025).

Neural Nexus Hackathon (2026), participant | Bounty Hunters Hackathon (2026), participant.